

Integrating FADCs

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I INTRODUCTION AND GENERAL DESCRIPTION

This is a specification for an integrating FADC to be used for data acquisition in Hall A at Jefferson Lab. The intention is to build a device that is a feasibly small change on an existing commercial design, with added features required for our program.

The data are fast PMTs signals from detectors typically -2 Volts (i.e. usually negative) and FWHM of order 50 nsec. FADCs are desirable to study in detail the structure of the data and to understand noise, background, and pileup effects at high rates. A pipelining feature can be used in the HRS DAQ to trigger the data and pick it out of a circular buffer, eliminating the need for analog delay lines. The commercial unit will be like a SISGmbH SIS3320 or a CAEN V1720 (the latter will be on the market in June 2007).

Another major goal of the DAQ is a digital integration of the data. A class of experiments require high precision measurements of signals integrated over a helicity pulse (33 msec), or over a fraction as small as $1/20$ of a helicity pulse, this latter case is called “oversampling”. The integration method has the potential to eliminate a host of systematic errors which have historically limited the precision of the results.

An overview of the required specs is in table 1. There is some flexibility in these specs, so if something is difficult here the vendor may discuss it with us. The main feature which is new are the so-called “accumulators”, which we will specify below. These accumulators provide digital integration with practically zero deadtime.

These FADCs may be used by the Hall A Compton polarimeter, the Deep Inelastic Parity program, and the 12 GeV upgrade DAQ for the Hall A Spectrometers. Initially, for evaluation, we will buy two FADC units with a delivery time of 5 months. We will need ~ 50 channels for the first two programs mentioned, in the the time frame of 2 years. The 12 GeV upgrade requires ~ 500 channels and is on a longer timescale of about 4 years.

II SPECIFICATION OF THE ACCUMULATORS

For each channel we want “accumulator” values which will be digital sums of FADC values within a gated time interval, subject to certain threshold conditions defined below. The time-gating can be done with with a logic transition (either ECL or NIM is fine) at the start and stop times T_{start} and T_{stop} . A typical value for the time interval $T_{\text{stop}} - T_{\text{start}}$ will be 30 msec. It is desirable but *not* required to have an option of providing a gate time that is programmable within the module for standalone tests.

For each of the following accumulators we need two pieces of information: 1) the accumulator itself and 2) the number of samples that went into that accumulator. Since the accumulators will be 36 bits it would be ok to spread its value over two 32-bit registers and we can build the 36-bit word later. But we need all the bits.

The accumulators are filled if they obey cut conditions defined by user-programmable thresholds. There are two of these thresholds, call them Thr_1 and Thr_2 . These are ADC values which the accumulator algorithms use as cuts. Some typical signals and how they align with the thrsholds are shown in fig 1.

The following defines the six accumulators. Note, the direction of the cut (“<” vs “>”) depends on how the FADC changes with voltage. Here it is assumed that, as for the SIS3320, the larger FADC values correspond to more positive voltages. Our signals are usually negative. If the FADC proposed by the vendor responds differently to voltages we’ll need to be clear about the cut definitions.

For Accumulators 5 and 6, we need user-programmable parameters to define

how long ahead and how long after the threshold to accumulate the data.

1. Accum1 = The digital sum between T_{start} and T_{stop} with *no* cut on the data.
2. Accum2 = The digital sum between T_{start} and T_{stop} for FADC values $> Thr_1$. This is a “low” threshold if we assume the aforementioned model for FADC response and the fact that our signals are negative – this cut causes accumulation when the data are near pedestal.
3. Accum3 = The digital sum between T_{start} and T_{stop} for FADC values that obey $Thr_2 < \text{FADC} \leq Thr_1$. The idea here is to accumulate for data between a “low” and “high” threshold (Thr_1 and Thr_2 , resp.).
4. Accum4 = The digital sum between T_{start} and T_{stop} for FADC values $\leq Thr_2$. This is a “high” threshold cut.
5. Accum5 = Data are summed between T_{start} and T_{stop} in the following way: Starting at the instant that the FADC is $\leq Thr_1$ (i.e. “above” the low threshold), the previous N_{before}^5 and the subsequent N_{after}^5 pulses are added to the accumulator. However, **data are not summed into Accum5 if they are summed into Accum6..** The variables N_{before}^5 and N_{after}^5 are programmable in the range between zero and 200. The time range T_{start} to T_{stop} over-rides the range N_{before}^5 to N_{after}^5 . Typical values would be $N_{\text{before}}^5 = 5$ and $N_{\text{after}}^5 = 20$.
6. Accum6 = Data are summed between T_{start} and T_{stop} in the following way: Starting at the instant that the FADC is $\leq Thr_2$ (i.e. “above” the high threshold), the previous N_{before}^6 and the subsequent N_{after}^6 pulses are added to the accumulator. The variables N_{before}^6 and N_{after}^6 are programmable in the range between zero and 200. The time range T_{start} to T_{stop} over-rides the range N_{before}^6 to N_{after}^6 . Typical values would be $N_{\text{before}}^6 = 6$ and $N_{\text{after}}^6 = 20$. Note that, as stated above, data that satisfies the Accum6 cut *do not* get added to Accum5.

We also need to know the number of samples that went into each of these accumulators, N_i^{samples} for $i = 1 - 6$. Note that for the first 4 accumulators

$$N_1^{\text{samples}} = N_2^{\text{samples}} + N_3^{\text{samples}} + N_4^{\text{samples}} \quad (1)$$

If there is something particularly difficult about this specification we can discuss it with the vendor. There is some flexibility about what we can accept.

The usage this information for data analysis is written in the appendix for our colleagues.

TABLE 1. ADC Specification

Quantity	Specification	Comments
Bit Resolution	12 bits	
Sampling Rate	250 MHz	
Dynamic Range	± 5 V	Offset adjustable with DAC
Input Polarity	Normally negative	Bipolar required
Buffer Depth	8.25 Msamples per chan	33 msec
Input Signals	PMT signals (0 to -5 V)	
Accumulator Feature	see text	

III APPENDIX: DATA ANALYSIS WITH ACCUMULATORS

This appendix is written mainly for our colleagues, so they may understand the intended use of the FADCs. It is not an essential part of the specification.

The triggerable feature described in the introduction is a standard feature of the commercial FADCs. We will be able to use them in the HRS DAQ to replace obsolete Fastbus 1881M, and for Hall A's purposes the commercial units should be competitive with the design which the JLab electronics group is planning to produce for Hall D. Evaluating this will be one purpose of buying some units for evaluation.

The analysis with the accumulator will be discussed next. The ADC values A are related to the input voltage V by

$$A = C_{\text{conv}}(V - V_0) \text{ for } V \geq V_0 \text{ ; and } A = 0 \text{ for } V \leq V_0 \quad (2)$$

where C_{conv} is a conversion and V_0 is the voltage at which the ADC reads zero. The voltages in our application are generally negative, and we are interested in the absolute value of the voltage, which for the SIS3320 ADC is:

$$|V| = \frac{1}{C_{\text{conv}}}(P - A) \quad (3)$$

where pedestal P is defined as the ADC value corresponding to zero voltage.

The accumulator is a sum over samples $i = 1, 2, 3, \dots, N^{\text{samples}}$ and is $\sum A_i$. Therefore the integrated signal S from T_{start} to T_{stop} is

$$S = \frac{1}{C_{\text{conv}}}(P \times N^{\text{samples}}) - \sum_i A_i \quad (4)$$

For computing the asymmetry between helicity periods, the conversion factor cancels. We will therefore drop C_{conv} in the following.

As an example for how to proceed with the data analysis we consider the physics signal corresponding to data above a low threshold (possibly necessary to reduce noise) and below a high threshold (set at the Compton edge) defined from Accum3:

$$S_3 = (P \times N_3^{\text{samples}}) - \text{Accum3} \quad (5)$$

where N_3^{samples} is the number of samples contributing to Accum3. Notice we need to know *both* Accum3 and N_3^{samples} to compute the physics signal. We can similarly define signals S_1 , S_2 , and S_4 in terms of the Accum1, Accum2, and Accum4.

For application to Compton polarimetry, a lower threshold is desirable to reduce noise. The upper threshold Thr_2 may be put just above the Compton edge to exclude high energy background due to Brehmstrahlung. The systematic error introduced by either of these thresholds must be demonstrated to be negligible.

The accumulator Accum4 introduces a deadtime correction “ ϵ ” to Accum3 since the time spent contributing to Accum4 was robbed from Accum3.

$$\epsilon = \frac{N_4^{\text{samples}}}{N_1^{\text{samples}}} \quad (6)$$

We compute a physics signal S_{phy} as a “corrected Accum3 signal”. This will be used to compute the physics asymmetry.

$$S_{\text{phy}} = \text{Accum3} \frac{1}{1 - \epsilon} (1 - \alpha \text{Accum4}) \quad (7)$$

where we have corrected for deadtime – the $1/(1-\epsilon)$ factor – and where α is an empirical correction factor for the high energy background due to the influence of the tails of the background on Accum3. We must rely on a average measurement of α using laser-off runs. This background correction is applied on a pulse-by-pulse basis which should be more accurate than applying it globally to an entire run. Systematic errors due in α and those caused by the fluctuations in the high energy tail will need to be evaluated. Information from the full-sampling mode will permit this evaluation.

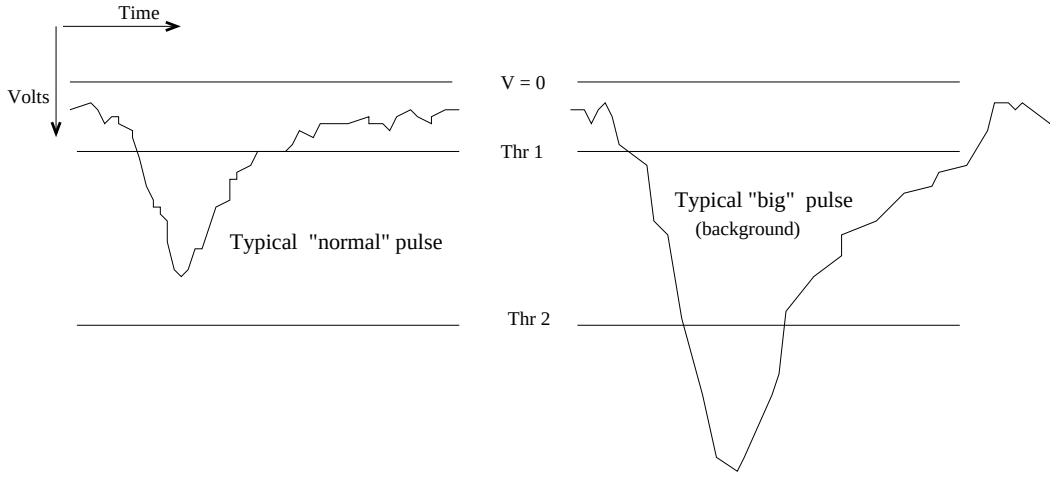


FIGURE 1. Typical small and big signals with the indicated thresholds.

Other analysis methods are possible using the accumulators. For example, we may find that the lower threshold is unnecessary. Then we could use $\text{Accum2} + \text{Accum3}$ to obtain the integral below the Compton edge; or equivalently set the Thr_1 to the high positive end of the ADC, effectively turning it off, then using Accum3 .

Accum5 and accum6 will allow us to count the signal above a noise floor (accum5) and the signal above the Compton edge (accum6). The latter is our background. The desirable feature of accum5 & 6 is that we count for a fixed time interval about a threshold. This will be better at integrating the tails and makes the system more linear than using a simple “count above threshold” scheme.

As a final comment, for future applications of FADCs it would be desirable to have flexible programming on the FPGA to compute items of physics interest from the data with minimal or no deadtime. A kind of “C to FPGA cross compiler” would be useful, although there would be limitations on the complexity of the algorithm so as not to cause deadtime.